

FORM 2
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COMPLETE SPECIFICATION
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AI Enabled Crop System for Autonomus Fruit Picker Robot

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The following specification particularly describes the invention and the manner in which it is to be performed.

AI Enabled Crop System for Autonomus Fruit Picker Robot

FIELD OF INVENTION

The field of invention is primarily relates to the field of agricultural automation and within the domains of Artificial Intelligence (AI), Machine Learning (ML), Computer Vision (CV) and Internet of Things (IoT).

BACKGROUND OF INVENTION

Conventional agricultural practices, particularly those related to harvesting and plant health monitoring, rely heavily on manual labor. This dependency presents multiple challenges including labor shortages, high operational costs, inconsistencies in crop handling, and delayed detection of plant diseases. Manual harvesting often leads to variability in fruit quality due to subjective judgment in assessing ripeness, and improper handling may result in physical damage to the produce. Additionally, early signs of plant diseases are frequently overlooked or detected too late, leading to reduced crop yield and potential spread of infections across large farming areas. With the advent of precision agriculture and advancements in robotics and artificial intelligence, there is a growing interest in automating key agricultural processes. Although various technologies exist for either harvesting or disease detection, there is a lack of integrated systems capable of performing both tasks simultaneously in a cost-effective and scalable manner. Existing robotic harvesting solutions are often crop-specific, limited in mobility, or unable to operate effectively in diverse farming conditions. Therefore, there exists a need for a comprehensive solution that can automate fruit and vegetable harvesting while also monitoring plant health in real time. The present invention addresses this gap by providing an AI-driven robotic system RoboCrop that combines image-based ripeness detection, disease identification, and autonomous harvesting mechanisms. This innovation aims to enhance operational efficiency, reduce crop loss, and support sustainable agricultural practices by enabling smart, automated, and precise farming.

SUMMARY OF INVENTION

The present invention discloses an intelligent robotic system, referred to as RoboCrop, designed to automate the dual tasks of harvesting ripe produce and detecting early-stage plant diseases. The system leverages a combination of artificial intelligence, computer vision, and precision robotics to perform these agricultural functions with minimal human intervention. RoboCrop comprises multiple integrated components. The system begins with a set of cameras and sensors (101) that capture real-time images of fruits, vegetables, and surrounding foliage. These images are transmitted to the image processing unit (102), where AI-based algorithms analyze color, texture, and shape to assess ripeness. The processed data is further evaluated by the ripeness detection model (103) and the disease detection module (104), which scans for visible anomalies such as spots, discoloration, or deformities on fruits and leaves. Decisions based on image analysis are handled by the decision control unit (105), which determines whether the crop should be harvested or flagged for further inspection. Upon confirming harvest readiness, signals are sent to the robotic arm controller (106), which activates a precise mechanical arm to perform non-destructive harvesting (108). Simultaneously, the mobility and navigation system (107) controls the robot's movement through farm rows via a mobile base (109).

The operation of the system follows a logical sequence as illustrated in the flowchart. The process begins with the system start (201) and proceeds to image capture (202). The captured images are analyzed in step 203, leading to a conditional check for ripeness (204). If the fruit is not ripe, the system advances to the next plant (205). If ripe, a subsequent check for disease symptoms (206) is performed. In cases of detected disease, the system flags the crop (207). If no disease is found, it proceeds to activate the robotic arm (208) and harvest the fruit (209). The harvested data is then stored and logged (210), the system moves to the next plant (211), and the cycle repeats until termination (212). RoboCrop presents a scalable, cost-effective solution for modern farming environments by integrating visual analysis, autonomous control, and robotic precision in a single deployable unit. This invention significantly improves harvesting accuracy, reduces dependency on manual labor, and enables early detection of plant diseases, thereby promoting smart and sustainable agriculture.

DETAIL DESCRIPTION

The invention discloses a comprehensive robotic system, RoboCrop, engineered for autonomous harvesting and plant health monitoring in agricultural environments. The system is a multi-module, AI-integrated platform combining imaging, analysis, actuation, and mobility. The operation begins with the camera and sensor unit (101), which comprises RGB cameras and proximity sensors for acquiring real-time visual and spatial data of crops. These sensors capture high-resolution images and send them to the image processing unit (102), which runs on embedded edge computing platforms such as NVIDIA Jetson Nano or Raspberry Pi. The unit performs real-time pre-processing including color segmentation, background subtraction, and noise filtering using OpenCV.

The ripeness detection model (103) evaluates image features such as surface texture, coloration, and size using trained convolutional neural networks. This determines the readiness of fruits or vegetables for harvest. Simultaneously, the disease detection module (104) performs image classification to detect visual symptoms of plant diseases. This module uses neural network models, trained on labeled datasets of infected and healthy leaves or fruits. Decisions from the ripeness and disease modules are processed by the decision control unit (105), which executes rule-based logic or decision tree algorithms. If a fruit is classified as ripe and free of disease, control commands are sent to the robotic arm controller (106), which governs the robotic manipulator responsible for harvesting. The robotic arm (108) is actuated using servo motors or pneumatic systems and is equipped with soft grippers or vacuum-based end-effectors to ensure non-destructive harvesting.

For mobility, RoboCrop is built upon a mobile base (109), which is controlled by the mobility and navigation system (107). This system uses GPS and ultrasonic sensors to autonomously navigate through the rows of crops. Simultaneous Localization and Mapping (SLAM) algorithms, implemented via ROS (Robot Operating System), are used for obstacle detection and path planning. The control sequence is outlined in the accompanying flowchart diagram. The operation begins with system start (201), followed by image capture (202). The captured image is analyzed (203), and ripeness is evaluated (204). If the fruit is not ripe, the system navigates to the next plant (205). If ripe, disease detection is initiated (206). If symptoms are detected, the plant is flagged (207) for farmer inspection. If no disease is present, the robotic arm is activated (208), and the harvesting action is executed (209). Data from the operation is stored (210) for traceability and analytics. The robot then moves to the next plant (211), and this cycle continues until the system shuts down (212).

All sensor modules and actuators communicate through standard protocols. The entire software stack is built using Python, TensorFlow, PyTorch, and ROS. The modular design allows the system to be reconfigured for different types of crops and varying field topologies, making RoboCrop highly scalable and adaptable for diverse agricultural scenarios.

DETAIL DESCRIPTION OF DRAWINGS

An exemplary embodiment of figure 1 is illustrated.

An exemplary embodiment of figure 2 is illustrated a flowchart depicting workflow.

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WE CLAIM

1. A RoboCrop system (100) comprising:
 - i. A vision and sensor module (101) configured to capture real-time images and environmental data using integrated RGB cameras, infrared sensors, and proximity detectors for assessing fruit ripeness and detecting visible plant diseases;
 - ii. An AI-based processing unit (102) including a ripeness detection model and a disease detection module, both trained on agricultural image datasets, and configured to analyze captured data for decision-making on harvesting or disease flagging; and
 - iii. A robotic actuation and mobility module (103) comprising a robotic arm with non-destructive harvesting capability and an autonomous navigation system based on SLAM, enabling the robot to move between plants and perform targeted picking operations.
2. A RoboCrop system (100) as claimed in claim 1, wherein the ripeness detection model (103) is trained using labeled crop datasets and is deployed on an edge computing platform capable of real-time inference;
3. A RoboCrop system (100) as claimed in claim 1, wherein the disease detection module (104) is trained to detect specific patterns of plant leaf damage, including but not limited to, spotting, blight, or mildew, using high-resolution imagery;
4. A RoboCrop system (100) as claimed in claim 1, wherein the robotic arm (108) is fixed to adapt to varying fruit sizes and orientations without damaging the plant or adjacent produce;
5. A RoboCrop system (100) as claimed in claim 1, wherein the mobility and navigation system (107) uses simultaneous localization and mapping (SLAM) algorithms for autonomous field traversal and positioning;
6. A RoboCrop system (100) as claimed in claim 1, wherein the integration of ripeness detection (103) and disease detection (104) is executed in parallel on a unified image stream, enabling immediate actuation decisions without requiring manual intervention.
7. A RoboCrop system (100) as claimed in claim 1, wherein harvested data including fruit type, location, ripeness status, and health records are logged and optionally transmitted wirelessly for cloud-based farm analytics; and
8. A RoboCrop system (100) is intended for use in modern agricultural operations for the autonomous harvesting of fruits and vegetables while simultaneously monitoring plant health, thereby reducing human labor dependency, improving harvest accuracy, and supporting scalable, AI-driven smart farming practices.

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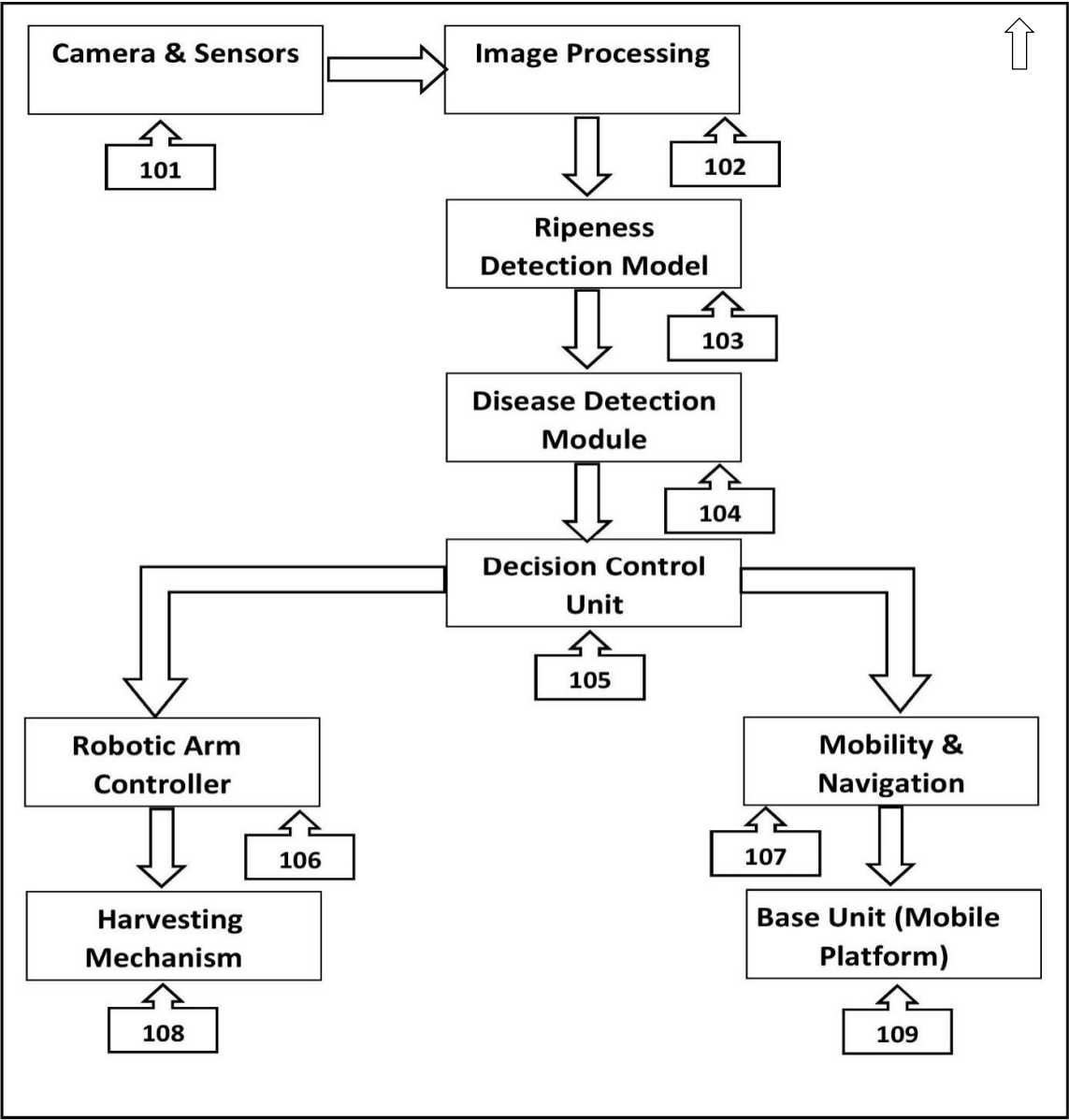


Fig 1

100

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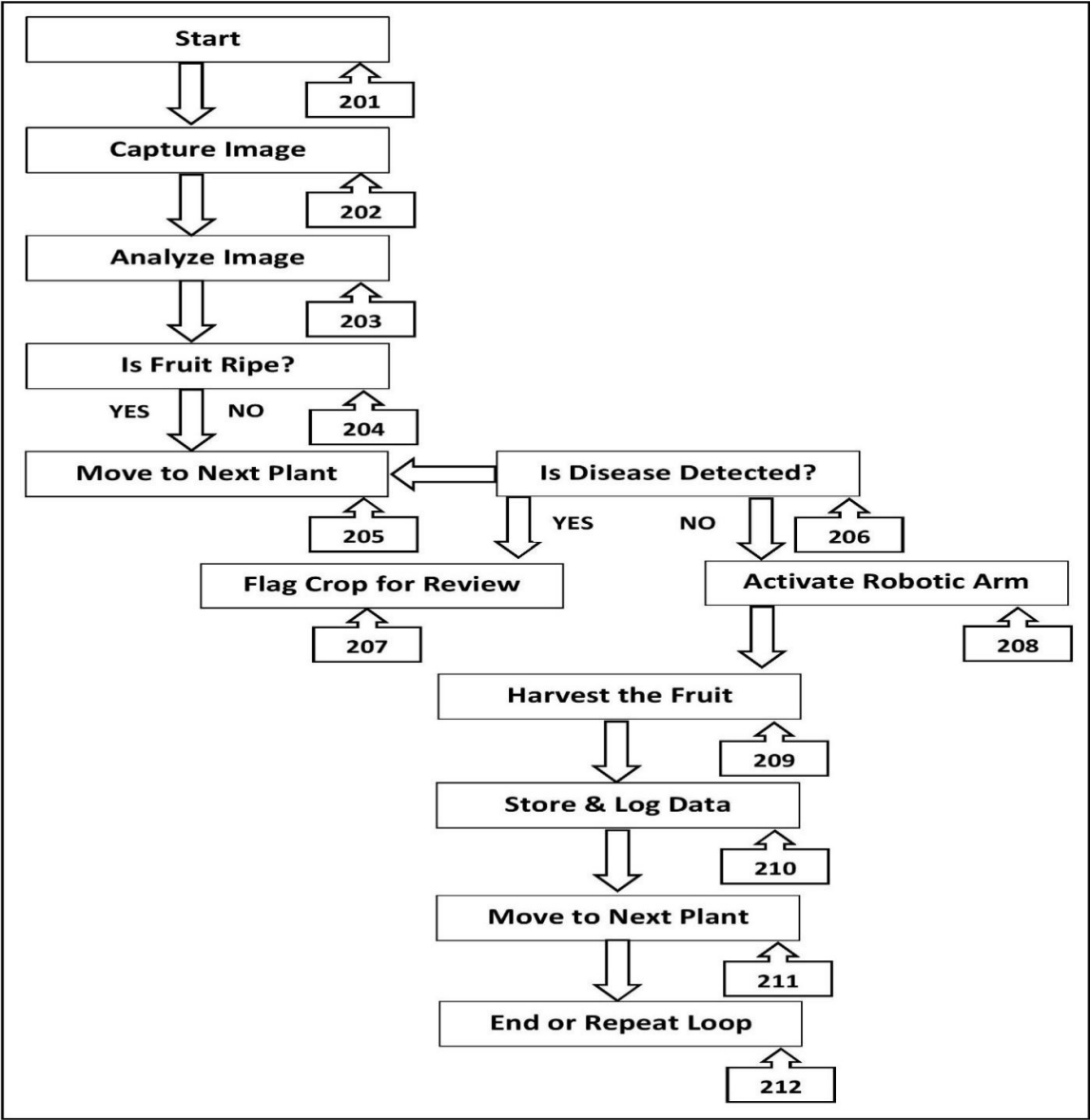


Fig 2

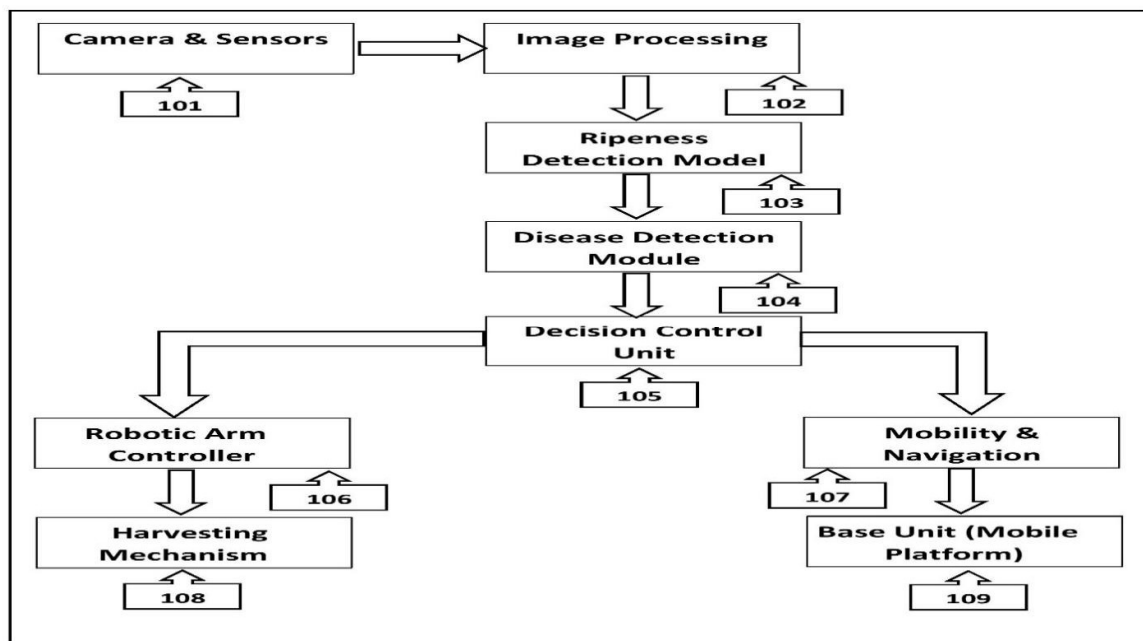
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ABSTRACT OF THE INVENTION

The present invention relates to a smart agricultural robot system (100) designed to automate the harvesting of fruits and vegetables while simultaneously monitoring plant health. The system comprises a vision and sensor module (101) that captures real-time images and environmental data using RGB cameras and proximity sensors; an AI-based processing unit (102) that includes a ripeness detection model and a disease detection module to analyze visual features and classify the condition of produce and plants; and a robotic actuation and mobility module (103) equipped with a multi-joint robotic arm for non-destructive harvesting and an autonomous navigation system based on SLAM techniques. The integration of modules (101), (102), and (103) enables RoboCrop to operate in dynamic farm environments, minimize labor dependency, ensure precise harvesting, and detect early signs of disease—thereby improving agricultural productivity and supporting smart farming practices.



↑
Fig 100

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